

Raj Pund

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Internship

Unified Mentor | Remote Internship

Feb 2025 - April 2025

Data Science Intern

- Built and evaluated **machine learning models** using **Python, Scikit-learn, and Pandas**, improving prediction accuracy by **18%** through feature engineering and hyperparameter tuning.
- Implemented **exploratory data analysis (EDA)** and statistical analysis on **large structured datasets**, identifying key patterns that improved model feature selection and increased performance by **15%**.
- Created **interactive data visualizations and dashboards** using **Matplotlib, Seaborn, and Power BI**, enabling stakeholders to monitor KPIs and make **data-driven decisions faster by 30%**.

Projects

InsightTube | LangChain, RAG, Vector database, OpenAI API

Jan 2026 - March 2026

- Developed an **AI-powered YouTube Video Q&A system** where users input a **video ID** and query video content using natural language, leveraging **RAG architecture with LangChain and OpenAI embeddings**.
- Built a **semantic retrieval pipeline** by converting video transcripts into **vector embeddings stored in a FAISS database**, enabling contextual question answering from **100+ video transcripts** with **<5 second response time**.
- Implemented **prompt engineering and context-aware retrieval**, achieving **~95% grounded responses** while minimizing hallucinations in AI-generated answers.
- GitHub Repository Link: <https://github.com/pundraj/Langchain>

AgroSphere | Python, Scikit-Learn, XG-Boost, Random Forest

Aug 2025 - Nov 2025

- Developed a crop yield prediction system using ML algorithms (Linear Regression, Random Forest, SVM, XGBoost) on 10,000+ agricultural records, achieving 92% accuracy with XGBoost through GridSearchCV tuning.
- Performed feature engineering on 15+ variables including soil, weather, and irrigation data, improving model performance by 28% and reducing RMSE by 35% using correlation analysis and feature selection.
- Optimized model for real-time predictions (<200ms inference time), enabling data-driven fertilizer optimization (20% waste reduction) and 30% improvement in harvest planning accuracy.
- Github Repository Link: <https://github.com/pundraj/AgroSphere>

Certificates

A Guide to Machine Learning with Data Science

July 2025

CipherSchools — Certificate Link

Complete Cpp Course

December 2024

Neocolab — Certificate Link

Technical Skills

Languages: Python, C, C++, JavaScript, HTML, CSS

Tools/Platforms: MySQL, MongoDB, GitHub, Git, VS Code, Vercel, Figma, Canva

Data Analytics/ML: Excel, Power BI, Pandas, NumPy, Matplotlib, Seaborn, LangChain, fastAPI, RAG

Core Concepts: DSA, DBMS, Operating Systems, Machine Learning, Data Analysis, Feature Engineering

Education

Lovely Professional University Punjab

2023 – Present

Computer Science and Engineering – 7.52

Jalandhar, Punjab

Shri Mahant Ramgiri Baba Jr College

2021 – 2023

Intermediate – 65%

Partur, Maharashtra

Anand Primary and secondary school

2020 – 2021

Matriculation - 85%

Partur, Maharashtra